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How to Build a Complete Q-Matrix for a Cognitively Diagnostic Test

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Abstract: The Q-matrix of a cognitively diagnostic test is said to be complete if it guarantees the identifiability of all possible proficiency classes among examinees. An incomplete Q-matrix causes examinees to be assigned to proficiency classes to which they do not belong. Completeness of the Q-matrix is therefore a key requirement of any cognitively diagnostic test. The importance of the completeness property of the Q-matrix of a test as a fundamental condition to guarantee a reliable estimate of an examinee's attribute profile has only recently been realized by researchers. In fact, inspection of extant assessments based on the cognitive diagnosis framework often revealed that, in hindsight, the Q-matrices used with these tests were not complete. Thus, the availability of rules for building a complete Q-matrix at the early stages of test development is perhaps at least as desirable as rules for identifying the completeness of a given Q-matrix. This article presents procedures for constructing Q-matrices that are complete. The famous Fraction-Subtraction test problems by K. K. Tatsuoka (1984) are used throughout for illustration.

Keywords: Q-matrix; Completeness; Cognitive Diagnosis; Diagnostic Classification Models (DCMs); General DCMs.

1. Introduction

In the past decade, cognitive diagnosis has emerged as a new paradigm of educational measurement that seeks to combine rigorous psychometric standards with the goals of formative assessment. (DiBello, Roussos, and

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Stout, 2007; Haberman and von Davier, 2007; Leighton and Gierl, 2007; Rupp, Templin, and Henson, 2010). The focus is on specific examinee skills, knowledge, and other cognitive characteristics needed to perform tasks that are typically chosen to assess student ability. The close ties of assessments based on the cognitive diagnosis framework to curriculum are intentional and deemed to inform instruction and enhance student learning. Cognitively diagnostic tests are tailored to evaluate students' mastery of the instructional content and to provide immediate feedback on their strengths and weaknesses in terms of skills learned and skills needing study. Consequently, models of cognitive diagnosis—commonly referred to as "Diagnostic Classification Models" (DCMs)—describe ability in a domain as a composite of specific discrete skills—often called "cognitive attributes"—each of which an examinee may or may not have mastered based on the evidence of tasks that he or she performs. Different attribute profiles define distinct classes of proficiency. Modeling assessment data within the cognitive diagnosis framework involves estimating the item parameters (required for a detailed evaluation of the properties of a test) and assigning examinees to proficiency classes—that is, estimating their individual attribute profiles.

Cognitively diagnostic tests use items that themselves are characterized by individual profiles specifying which attributes are required to respond correctly to an item. The entire set of these item-attribute associations forms the Q-matrix of a test (Tatsuoka, 1985). The Q-matrix must be complete, which means it must guarantee the identifiability of all possible proficiency classes among examinees (Chiu, Douglas, and Li, 2009). An incomplete Q-matrix causes examinees to be assigned to proficiency classes to which they do not belong. Completeness of the Q-matrix is therefore a key requirement of any cognitively diagnostic test. The importance of Q-completeness as a fundamental condition for the cognitive diagnosis of assessment data has only been recently acknowledged (e.g., Chiu, 2013; DeCarlo, 2012; Liu, Xu and Ying, 2012, 2013).

But completeness of the Q-matrix is often difficult to establish, especially, for tests with a large number of items involving many attributes. As an additional complication, completeness is not an intrinsic property of the Q-matrix, but can only be assessed in relation to a specific DCM supposed to underlie the data—that is, the Q-matrix of a given test can be complete for one model, but incomplete for another. An even more intricate situation arises if the test items do not conform to a single DCM, but to a mix of several DCMs.

Recently, Köhn and Chiu (2017) investigated the technical requirements and conditions of Q-completeness, and how to use them to determine whether a given Q-matrix is complete. Of equal importance, however, seems the question of how to use these results in practice for constructing a complete Q-matrix for an assessment—especially, when the test in question is based on a complex DCM and uses a realistic number of items involving multiple attributes.

In this article, a sequential procedure is presented for constructing Q-matrices that are guaranteed to be complete regardless of which DCM is chosen. Part I of this article briefly reviews definitions and key technical concepts of cognitive diagnosis and Q-completeness. Part II presents the method for constructing a complete Q-matrix in terms of a case study based on Tatsuoka's (1984) famous set of fraction-subtraction problems. The discussion concludes with some practical recommendations.

2. Part I: Definitions and Key Technical Concepts

2.1 Diagnostic Classification Models

Let Y_{ij} denote the observed response of examinee i, $i = 1, 2, \dots, N$, to binary item j, $j = 1, 2, \dots, J$. For the general latent class model, the conditional probability of examinee i in proficiency class $\mathcal{C}_m$ answering item j correctly is defined by the item response function (IRF), $P(Y_{ij} = 1 | i \in \mathcal{C}_m) = \pi_{jm}$, where π_{jm} is constant for item j across all members i in proficiency class $\mathcal{C}_m$, $m = 1, 2, \dots, M$. The Y_{ij} are assumed independent conditional on proficiency-class membership (local independence); no further restrictions are imposed on the relation between the latent variable—proficiency-class membership—and the observed item response.

DCMs, in contrast, are constrained latent class models such that proficiency-class membership—associated with mastery of a particular attribute set—determines the probability of a correct item response. Suppose that K latent binary attributes constitute a certain ability domain; there are then 2^K distinct attribute profiles composed of these K attributes representing $M = 2^K$ distinct proficiency classes. (Note that an attribute profile for a proficiency class can consist of all zeroes, because it is possible for an examinee not to have mastered any attributes at all.) Let the K-dimensional vector, $\boldsymbol{\alpha}_m = (\alpha_{m1}, \alpha_{m2}, \dots, \alpha_{mK})'$, be the binary attribute profile of proficiency class $\mathcal{C}_m$, where the k^{th} entry, $k = 1, 2, \dots, K$, indicates whether the corresponding attribute has been mastered. The attribute profile of examinee $i \in \mathcal{C}_m$, $\boldsymbol{\alpha}_{i \in \mathcal{C}_m}$, is usually simply written as $\boldsymbol{\alpha}_i$. (For brevity, the examinee index, i, is omitted if the context permits.)

Consider a test of J items for assessing ability in the domain. Each individual item j is associated with a K-dimensional binary vector $\mathbf{q}_j$ called the item-attribute profile, where $q_{jk} = 1$ if a correct answer requires mastery of the k^{th} attribute, and 0 otherwise. Note that item-attribute profiles consisting entirely of zeroes are inadmissible, because they correspond to items that require no attributes at all. Hence, given K attributes, there are at most $2^K - 1$ distinct item-attribute profiles. The item-attribute profiles of a test are collected into the $J \times K$ Q-matrix $\mathbf{Q} = \{q_{jk}\}$.

A plethora of DCMs has been proposed in the literature (e.g., Fu and Li, 2007; Rupp and Templin, 2008). They differ in how the functional relation between mastery of attributes and the probability of a correct item response is modeled. DCMs have been distinguished based on criteria like compensatory versus non-compensatory (can lacking certain attributes be compensated for by possessing other attributes or not), or conjunctive (all attributes specified for an item in $\mathbf{q}$ are required; mastering only a subset of them results in a success probability equal to that of an examinee mastering none of the attributes) versus disjunctive (mastery of a subset of the required attributes is a sufficient condition for maximizing the probability of a correct item response) (de la Torre and Douglas, 2004; Henson, Templin, and Willse, 2009; Maris, 1999). The Deterministic Input Noisy "AND" Gate (DINA) Model (Haertel, 1989; Junker and Sijtsma, 2001; Macready and Dayton, 1977) is the standard example of a conjunctive DCM. Its IRF is $P(Y_{ij} = 1 \mid \boldsymbol{\alpha}_i) = (1 - s_j)^{\eta_{ij}} g_j^{(1-\eta_{ij})}$, subject to $0 < g_j < 1 - s_j < 1$ for each item j. The conjunction parameter η_{ij} is defined as $\eta_{ij} = \prod_{k=1}^{K} \alpha_{ik}^{q_{jk}}$ indicating whether examinee i has mastered all the attributes needed to answer item j correctly. The item-related parameters $s_j = P(Y_{ij} = 0 \mid \eta_{ij} = 1)$ and $g_j = P(Y_{ij} = 1 \mid \eta_{ij} = 0)$ refer to the probabilities of slipping (failing to answer item j correctly despite having the attributes required to do so) and guessing (answering item j correctly despite lacking the attributes required to do so), respectively. The Deterministic Input Noisy "OR" Gate (DINO) Model (Templin and Henson, 2006) is the prototypical disjunctive DCM. The disjunction parameter, $\omega_{ij} = 1 - \prod_{k=1}^{K}(1 - \alpha_{ik})^{q_{jk}}$, indicates whether at least one of the attributes associated with item j has been mastered. (Like η_{ij} in the DINA model, ω_{ij} represents the ideal item response when neither slipping nor guessing occur.) The IRF of the DINO model is $P(Y_{ij} = 1 \mid \boldsymbol{\alpha}_i) = (1 - s_j)^{\omega_{ij}} g_j^{(1-\omega_{ij})}$, subject to $0 < g_j < 1 - s_j < 1$ for each item j.

General DCMs have been proposed as a theoretical framework for expressing the IRFs of specific DCMs in unified mathematical form and parameterization (de la Torre, 2011; Henson et al., 2009; Rupp et al., 2010; von Davier, 2005, 2008, 2014). Von Davier's General Diagnostic Model (GDM; 2005, 2008) is the archetypal general DCM. The IRF of (presumably) the most popular version of the GDM is formed by the logistic function of the linear combination of all K attribute main effects. Henson et al. (2009) proposed to use the linear combination of the K attribute main effects and all their two-way, three-way, ..., K-way interactions

$$
\begin{aligned}
v_{ij} &= \beta_{j0} + \sum_{k=1}^{K} \beta_{jk} q_{jk} \alpha_{ik} + \sum_{k'=k+1}^{K} \sum_{k=1}^{K-1} \beta_{j(kk')} q_{jk} q_{jk'} \alpha_{ik} \alpha_{ik'} + \\
&\quad \cdots + \beta_{j12\ldots K} \prod_{k=1}^{K} q_{jk} \alpha_{ik} \qquad (1)
\end{aligned}
$$

(q_{jk} indicates whether mastery of α_k is required for item j) for constructing the IRF of a general DCM called the Loglinear Cognitive Diagnosis Model (LCDM)

$$
P(Y_{ij} = 1 \mid \boldsymbol{\alpha}_i) = \frac{\exp(v_{ij})}{1 + \exp(v_{ij})} \qquad (2)
$$

(see Equation 11 in Henson et al., 2009). De la Torre (2011) presented the Generalized DINA (G-DINA) model that, in addition to the logit link, allows for constructing the IRF based on the identity link, $P(Y_{ij} = 1 \mid \boldsymbol{\alpha}_i) = v_{ij}$, and the log link, $P(Y_{ij} = 1 \mid \boldsymbol{\alpha}_i) = \exp\left(v_{ij}\right)$ (see Equations 1–3 in de la Torre, 2011). By imposing appropriate constraints on the β-coefficients in v_{ij}, the IRFs of specific DCMs can be reparameterized as general DCMs.

For example, the IRF of the DINA model becomes

$$
P(Y_{ij} = 1 \mid \boldsymbol{\alpha}_i) = \frac{\exp\left(\beta_{j0} + \beta_{j(\forall k \in \mathcal{L}_j)} \prod_{k \in \mathcal{L}_j} \alpha_{ik}\right)}{1 + \exp\left(\beta_{j0} + \beta_{j(\forall k \in \mathcal{L}_j)} \prod_{k \in \mathcal{L}_j} \alpha_{ik}\right)} \qquad (3)
$$

subject to $\beta_{j(\forall k \in \mathcal{L}_j)} > 0$. The set $\mathcal{L}_j = \{k \mid q_{jk} = 1\}$ contains the non-zero elements in $\mathbf{q}_j$. (If $k \in \mathcal{L}_j$, then $q_{jk} = 1$ is always true; hence, q_{jk} has been dropped from the IRF.) Henceforth, all DCMs are written as general DCMs.

2.2 Completeness of the Q-Matrix

2.2.1 Definition

Chiu et al. (2009) proposed a formal definition of Q-completeness in terms of the vectors of ideal item responses, $\boldsymbol{\eta}(\boldsymbol{\alpha}) = \boldsymbol{\eta}(\boldsymbol{\alpha}^*) \Rightarrow \boldsymbol{\alpha} = \boldsymbol{\alpha}^*$, with entries η_j as defined earlier. Verbally stated, a $J \times K$ matrix $\mathbf{Q} = \{q_{jk}\}$ is said to be complete if it allows for the identification of all the possible attribute profiles of the $2^K = M$ different proficiency classes. This definition was tailored to the DINA model, but it is not applicable to DCMs without conjunction parameter. Thus, Chiu and Köhn (2015) proposed an equivalent, yet more general definition of completeness of the Q-matrix based on the conditional expectation of an examinee's J-dimensional item response profile $\boldsymbol{Y}$, given attribute profile $\boldsymbol{\alpha}$, denoted by $\boldsymbol{S}(\boldsymbol{\alpha}) = E(\boldsymbol{Y} \mid \boldsymbol{\alpha})$. (For the DINA model, the j^{th} entry of $\boldsymbol{S}(\boldsymbol{\alpha})$ is defined as $E(Y_j \mid \boldsymbol{\alpha}) = (1 - s_j)^{\eta_j} g_j^{(1-\eta_j)}$.)—Here is a small-scale example.

The Q-matrix $\mathbf{Q}^{(1:3)}$ has three rows that are the attribute profiles $\mathbf{q}_1$, $\mathbf{q}_2$, and $\mathbf{q}_3$ for $J = 3$ items; the $K = 3$ columns correspond to the attributes α_1, α_2, and α_3. (The matrix superscript refers to the item indices $j = 1, 2, 3$.)

$$\mathbf{Q}^{(1:3)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Note that each item requires only a single attribute. Completeness of $\mathbf{Q}^{(1:3)}$ is established by checking the definition $\boldsymbol{S}(\boldsymbol{\alpha}) = \boldsymbol{S}(\boldsymbol{\alpha}^*) \Rightarrow \boldsymbol{\alpha} = \boldsymbol{\alpha}^*$ for each pair, $\boldsymbol{\alpha}$ and $\boldsymbol{\alpha}^*$, of the $M = 8$ proficiency classes. The elements of the vector of expected item responses, $\boldsymbol{S}(\boldsymbol{\alpha}) = \big(S_1(\boldsymbol{\alpha}), S_2(\boldsymbol{\alpha}), S_3(\boldsymbol{\alpha})\big)'$, are defined as $S_j(\boldsymbol{\alpha}) = E(Y_j \mid \boldsymbol{\alpha}) = P(Y_j \mid \boldsymbol{\alpha})$. (Recall that the random variable Y_j is distributed as a Bernoulli(π_j), where $\pi_j = P(Y_j = 1 \mid \boldsymbol{\alpha})$, as defined by the IRF of the DCM in question; hence, $E(Y_j \mid \boldsymbol{\alpha}) = P(Y_j = 1 \mid \boldsymbol{\alpha})$.) As a simple illustration, consider a main-effects-only model like the GDM:

$$S_j(\boldsymbol{\alpha}) = \frac{\exp\left(\beta_{j0} + \sum_{k=1}^{K} \beta_{jk} q_{jk} \alpha_k\right)}{1 + \exp\left(\beta_{j0} + \sum_{k=1}^{K} \beta_{jk} q_{jk} \alpha_k\right)}.$$

If $\mathbf{q}_1 = (100)$, then for

$$\begin{aligned} \boldsymbol{\alpha} = (000)', \qquad & S_1\big((000)'\big) = \tfrac{\exp(\beta_{10})}{1+\exp(\beta_{10})} \\ \boldsymbol{\alpha} = (100)', \qquad & S_1\big((100)'\big) = \tfrac{\exp(\beta_{10}+\beta_{11}\alpha_1)}{1+\exp(\beta_{10}+\beta_{11}\alpha_1)}. \end{aligned}$$

$\mathbf{Q}^{(1:3)}$ allows to distinguish all vectors $\boldsymbol{S}(\boldsymbol{\alpha})$ of expected item responses associated with different attribute profiles $\boldsymbol{\alpha}$; thus, all proficiency classes are identifiable. Note that only the coefficients retained in $S_j(\boldsymbol{\alpha})$ are reported in the table below, but not the expression of the entire logistic function:

$\boldsymbol{\alpha} = (\alpha_1\ \alpha_2\ \alpha_3)'$	$\mathbf{Q}^{(1:3)}$		
	$\mathbf{q}_1 = (100)$	$\mathbf{q}_2 = (010)$	$\mathbf{q}_3 = (001)$
	$S_1(\boldsymbol{\alpha})$	$S_2(\boldsymbol{\alpha})$	$S_3(\boldsymbol{\alpha})$
$\boldsymbol{\alpha}_1 = (000)'$	β_{10}	β_{20}	β_{30}
$\boldsymbol{\alpha}_2 = (001)'$	β_{10}	β_{20}	$\beta_{30} + \beta_{33}$
$\boldsymbol{\alpha}_3 = (010)'$	β_{10}	$\beta_{20} + \beta_{22}$	β_{30}
$\boldsymbol{\alpha}_4 = (011)'$	β_{10}	$\beta_{20} + \beta_{22}$	$\beta_{30} + \beta_{33}$
$\boldsymbol{\alpha}_5 = (100)'$	$\beta_{10} + \beta_{11}$	β_{20}	β_{30}
$\boldsymbol{\alpha}_6 = (101)'$	$\beta_{10} + \beta_{11}$	β_{20}	$\beta_{30} + \beta_{33}$
$\boldsymbol{\alpha}_7 = (110)'$	$\beta_{10} + \beta_{11}$	$\beta_{20} + \beta_{22}$	β_{30}
$\boldsymbol{\alpha}_8 = (111)'$	$\beta_{10} + \beta_{11}$	$\beta_{20} + \beta_{22}$	$\beta_{30} + \beta_{33}$

Compare these results with those obtained for the Q-matrix $\mathbf{Q}^{(1:2)}$

$$\mathbf{Q}^{(1:2)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$\boldsymbol{\alpha} = (\alpha_1\ \alpha_2\ \alpha_3)'$	$\mathbf{Q}^{(1:2)}$	
	$\mathbf{q}_1 = (100)$	$\mathbf{q}_2 = (010)$
	$S_1(\boldsymbol{\alpha})$	$S_2(\boldsymbol{\alpha})$
$\boldsymbol{\alpha}_1 = (000)'$	β_{10}	β_{20}
$\boldsymbol{\alpha}_2 = (001)'$	β_{10}	β_{20}
$\boldsymbol{\alpha}_3 = (010)'$	β_{10}	$\beta_{20} + \beta_{21}$
$\boldsymbol{\alpha}_4 = (011)'$	β_{10}	$\beta_{20} + \beta_{21}$
$\boldsymbol{\alpha}_5 = (100)'$	$\beta_{10} + \beta_{11}$	β_{20}
$\boldsymbol{\alpha}_6 = (101)'$	$\beta_{10} + \beta_{11}$	β_{20}
$\boldsymbol{\alpha}_7 = (110)'$	$\beta_{10} + \beta_{11}$	$\beta_{20} + \beta_{21}$
$\boldsymbol{\alpha}_8 = (111)'$	$\beta_{10} + \beta_{11}$	$\beta_{20} + \beta_{21}$

Clearly, $\mathbf{Q}^{(1:2)}$ is not complete because only four proficiency classes can be identified. For example, $\boldsymbol{S}(\boldsymbol{\alpha}_1) = \boldsymbol{S}(\boldsymbol{\alpha}_2)$, whereas $\boldsymbol{\alpha}_1 = (000)' \neq \boldsymbol{\alpha}_2 = (001)'$. Likewise, $\boldsymbol{S}(\boldsymbol{\alpha}_3) = \boldsymbol{S}(\boldsymbol{\alpha}_4)$, but $\boldsymbol{\alpha}_3 = (010)' \neq \boldsymbol{\alpha}_4 = (011)'$; $\boldsymbol{S}(\boldsymbol{\alpha}_5) = \boldsymbol{S}(\boldsymbol{\alpha}_6)$, but $\boldsymbol{\alpha}_5 = (100)' \neq \boldsymbol{\alpha}_6 = (101)'$; and $\boldsymbol{S}(\boldsymbol{\alpha}_7) = \boldsymbol{S}(\boldsymbol{\alpha}_8)$, but $\boldsymbol{\alpha}_7 = (110)' \neq \boldsymbol{\alpha}_8 = (111)'$. Hence, $\mathbf{Q}^{(1:2)}$ does not allow to distinguish all $M = 8$ proficiency classes—which is also suggested by intuition because two items can only result in four distinct manifest response patterns, (00), (01), (10), and (11), where the first number corresponds to the response to item $Y_1 = 0, 1$; the second to $Y_2 = 0, 1$. Thus, for the proper discrimination of all proficiency classes, a third item is required.

2.3 Rules of Q-Completeness

From the theorems and propositions presented in Köhn and Chiu (2017), three rules can be derived to describe the technical conditions of Q-completeness.

2.3.1 Rule 1

A $J \times K$ Q-matrix is guaranteed to be complete for any DCM if it contains among its J rows the K single-attribute items $\mathbf{e}_1, \ldots, \mathbf{e}_K$.

($\mathbf{e}_k$ denotes a $1 \times K$ vector, with the k^{th} element, e_k, equal to 1, and all other entries equal to 0.) Consider, for example, DCMs as different as the DINA model and the LCDM. For the DINA model,

$$S_j(\boldsymbol{\alpha}) = \frac{\exp\left(\beta_{j0} + \beta_{j(\forall k \in \mathcal{L}_j)} \prod_{k \in \mathcal{L}_j} \alpha_k\right)}{1 + \exp\left(\beta_{j0} + \beta_{j(\forall k \in \mathcal{L}_j)} \prod_{k \in \mathcal{L}_j} \alpha_k\right)}$$

(see Equation 3). Recall that the LCDM contains all attribute main and interaction effects (see Equations 1 and 2); hence,

$$\begin{aligned} S_j(\boldsymbol{\alpha}) = & \\ & \Bigg[\exp\Big(\beta_{j0} + \sum_{k=1}^{K} \beta_{jk} q_{jk} \alpha_k + \sum_{k'=k+1}^{K} \sum_{k=1}^{K-1} \beta_{j(kk')} q_{jk} q_{jk'} \alpha_k \alpha_{k'} + \\ & \quad \cdots + \beta_{j(12\ldots K)} \prod_{k=1}^{K} q_{jk} \alpha_k\Big)\Bigg] \Bigg/ \\ & \Bigg[1 + \exp\Big(\beta_{j0} + \sum_{k=1}^{K} \beta_{jk} q_{jk} \alpha_k + \sum_{k'=k+1}^{K} \sum_{k=1}^{K-1} \beta_{j(kk')} q_{jk} q_{jk'} \alpha_k \alpha_{k'} + \\ & \quad \cdots + \beta_{j(12\ldots K)} \prod_{k=1}^{K} q_{jk} \alpha_k\Big)\Bigg]. \end{aligned}$$

$\mathbf{Q}^{(1:3)}$ is complete for both models, as can be confirmed by computing the expected item response profiles, $\boldsymbol{S}(\boldsymbol{\alpha})$. Note that in case of single-attribute items, all interaction terms disappear from the $S_j(\boldsymbol{\alpha})$ of the LCDM, whereas for the DINA model the term $\prod_{k \in \mathcal{L}_j} \alpha_k$ is reduced to a simple main effect. Hence, for the DINA model and the LCDM, the expected item responses turn out to be identical to those calculated earlier for the GDM:

$\boldsymbol{\alpha} = (\alpha_1\ \alpha_2\ \alpha_3)'$	$\mathbf{Q}^{(1:3)}$		
	$\mathbf{q}_1 = (100)$	$\mathbf{q}_2 = (010)$	$\mathbf{q}_3 = (001)$
	$S_1(\boldsymbol{\alpha})$	$S_2(\boldsymbol{\alpha})$	$S_3(\boldsymbol{\alpha})$
$\boldsymbol{\alpha}_1 = (000)'$	β_{10}	β_{20}	β_{30}
$\boldsymbol{\alpha}_2 = (001)'$	β_{10}	β_{20}	$\beta_{30} + \beta_{33}$
$\boldsymbol{\alpha}_3 = (010)'$	β_{10}	$\beta_{20} + \beta_{22}$	β_{30}
$\boldsymbol{\alpha}_4 = (011)'$	β_{10}	$\beta_{20} + \beta_{22}$	$\beta_{30} + \beta_{33}$
$\boldsymbol{\alpha}_5 = (100)'$	$\beta_{10} + \beta_{11}$	β_{20}	β_{30}
$\boldsymbol{\alpha}_6 = (101)'$	$\beta_{10} + \beta_{11}$	β_{20}	$\beta_{30} + \beta_{33}$
$\boldsymbol{\alpha}_7 = (110)'$	$\beta_{10} + \beta_{11}$	$\beta_{20} + \beta_{22}$	β_{30}
$\boldsymbol{\alpha}_8 = (111)'$	$\beta_{10} + \beta_{11}$	$\beta_{20} + \beta_{22}$	$\beta_{30} + \beta_{33}$

For the DINA model and the DINO model, a $J \times K$ Q-matrix is complete if and only if each attribute is represented by at least one single-attribute item—this is a necessary condition of Q-completeness that only applies to those two DCMs, and that sets them apart from all other DCMs as special cases (see also, Chiu et al., 2009; Chiu and Köhn, 2015). As an illustration, consider $\mathbf{Q}^{(4:6)}$ that does not contain single-attribute items:

$$\mathbf{Q}^{(4:6)} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Computing the expected item responses shows that $\mathbf{Q}^{(4:6)}$ is not complete for the DINA model:

$\boldsymbol{\alpha} = (\alpha_1\ \alpha_2\ \alpha_3)'$	$\mathbf{Q}^{(4:6)}$		
	$\mathbf{q}_4 = (011)$	$\mathbf{q}_5 = (101)$	$\mathbf{q}_6 = (110)$
	$S_4(\boldsymbol{\alpha})$	$S_5(\boldsymbol{\alpha})$	$S_6(\boldsymbol{\alpha})$
$\boldsymbol{\alpha}_1 = (000)'$	β_{40}	β_{50}	β_{60}
$\boldsymbol{\alpha}_2 = (001)'$	β_{40}	β_{50}	β_{60}
$\boldsymbol{\alpha}_3 = (010)'$	β_{40}	β_{50}	β_{60}
$\boldsymbol{\alpha}_4 = (011)'$	$\beta_{40} + \beta_{4(23)}$	β_{50}	β_{60}
$\boldsymbol{\alpha}_5 = (100)'$	β_{40}	β_{50}	β_{60}
$\boldsymbol{\alpha}_6 = (101)'$	β_{40}	$\beta_{50} + \beta_{5(13)}$	β_{60}
$\boldsymbol{\alpha}_7 = (110)'$	β_{40}	β_{50}	$\beta_{60} + \beta_{6(12)}$
$\boldsymbol{\alpha}_8 = (111)'$	$\beta_{40} + \beta_{4(23)}$	$\beta_{50} + \beta_{5(13)}$	$\beta_{60} + \beta_{6(12)}$

However, for other DCMs, the inclusion of all K single-attribute items in the Q-matrix is a sufficient, but not a necessary condition of completeness, which means that compositions of the Q-matrix that do not include all single-attribute items can also be complete. Thus, $\mathbf{Q}^{(4:6)}$, for example, is complete for the LCDM despite the absence of single-attribute items:

$\boldsymbol{\alpha} = (\alpha_1\alpha_2\alpha_3)'$	$\mathbf{Q}^{(4:6)}$		
	$\mathbf{q}_4 = (011)$	$\mathbf{q}_5 = (101)$	$\mathbf{q}_6 = (110)$
	$S_4(\boldsymbol{\alpha})$	$S_5(\boldsymbol{\alpha})$	$S_6(\boldsymbol{\alpha})$
$\boldsymbol{\alpha}_1 = (000)'$	β_{40}	β_{50}	β_{60}
$\boldsymbol{\alpha}_2 = (001)'$	$\beta_{40} + \beta_{43}$	$\beta_{50} + \beta_{53}$	β_{60}
$\boldsymbol{\alpha}_3 = (010)'$	$\beta_{40} + \beta_{42}$	β_{50}	$\beta_{60} + \beta_{62}$
$\boldsymbol{\alpha}_4 = (011)'$	$\beta_{40} + \beta_{42} + \beta_{43} + \beta_{4(23)}$	$\beta_{50} + \beta_{53}$	$\beta_{60} + \beta_{62}$
$\boldsymbol{\alpha}_5 = (100)'$	β_{40}	$\beta_{50} + \beta_{51}$	$\beta_{60} + \beta_{61}$
$\boldsymbol{\alpha}_6 = (101)'$	$\beta_{40} + \beta_{43}$	$\beta_{50} + \beta_{51} + \beta_{53} + \beta_{5(13)}$	$\beta_{60} + \beta_{61}$
$\boldsymbol{\alpha}_7 = (110)'$	$\beta_{40} + \beta_{42}$	$\beta_{50} + \beta_{51}$	$\beta_{60} + \beta_{61} + \beta_{62} + \beta_{6(12)}$
$\boldsymbol{\alpha}_8 = (111)'$	$\beta_{40} + \beta_{42} + \beta_{43} + \beta_{4(23)}$	$\beta_{50} + \beta_{51} + \beta_{53} + \beta_{5(13)}$	$\beta_{60} + \beta_{61} + \beta_{62} + \beta_{6(12)}$

Note that this result also holds for a main-effects-only model like the GDM—then, the β-coefficients of the interaction terms, $\beta_{j(kk')}$, would disappear from the previous table, but $\boldsymbol{\alpha} \neq \boldsymbol{\alpha}^* \Rightarrow \boldsymbol{S}(\boldsymbol{\alpha}) \neq \boldsymbol{S}(\boldsymbol{\alpha}^*)$ would still hold for all pairs of the M proficiency classes.

These observations can be formalized as Rule 2, which expresses the sufficiency condition of Q-completeness in more general terms based on the concept of the rank of $\mathbf{Q}$. Recall from linear algebra that for an arbitrary matrix $\mathbf{A}$, with U rows and V columns, the maximum number of linearly independent rows and the maximum number of linearly independent columns is the same; this common number is defined to be the *rank* of $\mathbf{A}$. A matrix is said to be of *full rank* if the rank is equal to the minimum of U and V. 2.3.2

Rule 2

If the $J \times K$ matrix $\mathbf{Q}$ of a test is not of full rank K—that is, $\mathbf{Q}$ does not contain K linearly independent q-vectors—then, $\mathbf{Q}$ is not complete.

This rule applies independent of whether the DCM underlying the test is a main-effects-only model or a model with main and interaction effects. Within this context recall the examples of $\mathbf{Q}^{(1:3)}$ and $\mathbf{Q}^{(1:2)}$: the latter was not of full rank and thus, incomplete, whereas the full rank matrix $\mathbf{Q}^{(1:3)}$ was complete. However, having full rank K is a necessary, but not a sufficient condition of completeness—that is, a Q-matrix can be of full rank K, but still incomplete. Here is an example.

$$\mathbf{Q}^{(7:9)} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Note that $\mathbf{Q}^{(7:9)}$ has full rank. Computing the expected responses $\boldsymbol{S}(\boldsymbol{\alpha})$ and $\boldsymbol{S}(\boldsymbol{\alpha}^*)$ for the proficiency classes $\boldsymbol{\alpha} = (100)'$ and $\boldsymbol{\alpha}^* = (011)'$ shows $\boldsymbol{S}(\boldsymbol{\alpha})$ and $\boldsymbol{S}(\boldsymbol{\alpha}^*)$ cannot be guaranteed to be distinct unless the additional constraints $\beta_{71} \neq \beta_{72}$, $\beta_{81} \neq \beta_{83}$ or $\beta_{91} \neq \beta_{92} + \beta_{93}$ are imposed on the coefficients. Hence, $\mathbf{Q}^{(7:9)}$ may or may not be complete depending on the specific values of the β-coefficients:

$\boldsymbol{\alpha} = (\alpha_1\ \alpha_2\ \alpha_3)'$	$\mathbf{Q}^{(7:9)}$					
	$\mathbf{q}_7 = (110)$		$\mathbf{q}_8 = (101)$		$\mathbf{q}_9 = (111)$	
	$S_7(\boldsymbol{\alpha})$		$S_8(\boldsymbol{\alpha})$		$S_9(\boldsymbol{\alpha})$	
$\boldsymbol{\alpha} = (100)'$	$\beta_{70} + \beta_{71}$		$\beta_{80} + \beta_{81}$		$\beta_{90} + \beta_{91}$	
$\boldsymbol{\alpha}^* = (011)'$	β_{70}	$+\beta_{72}$	β_{80}	$+\beta_{83}$	β_{90}	$+\beta_{92} + \beta_{93}$

This example suggests that any candidate Q-matrix that is of full rank must be further inspected for completeness by checking all pairs of attribute profiles, $\boldsymbol{\alpha}$ and $\boldsymbol{\alpha}^*$, for possibly indeterminate β-coefficient patterns like β_{71} and β_{72}, β_{81} and β_{83}, or β_{91} and $\beta_{92} + \beta_{93}$ in the previous illustration. However, the amount of computational effort of checking for these indeterminacies explodes with the number K of attributes: for $K = 3$ attributes, $\binom{8}{2} = 56$ pairs must be checked, whereas if $K = 5$, $\binom{32}{2} = 496$ pairs must be checked. So, what to do?—The following result allows for a significant reduction in the computational effort: if $\boldsymbol{\alpha}$ and $\boldsymbol{\alpha}^*$ are *nested* within each other, then $\boldsymbol{S}(\boldsymbol{\alpha}) \neq \boldsymbol{S}(\boldsymbol{\alpha}^*)$ is always true. (A K-dimensional vector $\boldsymbol{\alpha}^* \neq \boldsymbol{\alpha}$ is said to be *nested* within the vector $\boldsymbol{\alpha}$ if $\alpha_k^* \leq \alpha_k$, for all elements k, and $\alpha_k^* < \alpha_k$ for at least one k.) Said differently, if $\boldsymbol{\alpha}$ and $\boldsymbol{\alpha}^*$ are not nested within each other, then indeterminate constellations of the β-coefficients in $\boldsymbol{S}(\boldsymbol{\alpha})$ and $\boldsymbol{S}(\boldsymbol{\alpha}^*)$ can occur like, for example, with $\boldsymbol{\alpha} = (100)'$ and $\boldsymbol{\alpha}^* = (011)'$, which might render the Q-matrix incomplete.

2.4 Summary: Checking Q-Completeness

Rules 1–2 together with the previous result suggest a sequential procedure for assessing the completeness of any given Q-matrix:

(1) Check wether $\mathbf{Q}$ contains all K single-attribute items. If so, then $\mathbf{Q}$ is complete.

(2) If any of the K single-attribute items is missing from $\mathbf{Q}$, and the data are fitted by the DINA model or the DINO model, then $\mathbf{Q}$ is not complete. If the fitted model is neither the DINA model nor the DINO model, then continue with (3).

(3) If the rank of $\mathbf{Q}$ is less than K, then $\mathbf{Q}$ is incomplete. If $\mathbf{Q}$ has full rank K, then continue with (4).

(4) Consider only all non-nested pairs $\boldsymbol{\alpha}$ and $\boldsymbol{\alpha}^*$ and check for possibly non-distinct $\boldsymbol{S}(\boldsymbol{\alpha})$ and $\boldsymbol{S}(\boldsymbol{\alpha}^*)$—concretely, if for each pair $\boldsymbol{\alpha}$ and $\boldsymbol{\alpha}^*$, there exists at least one item j such that the coefficients in $S_j(\boldsymbol{\alpha}^*)$ are a proper subset of those in $S_j(\boldsymbol{\alpha})$, or vice versa, then $\mathbf{Q}$ is complete; otherwise, $\mathbf{Q}$ is incomplete.

3. Part II: How to Build a Complete Q-Matrix—A Case Study

Recall that the importance of the completeness property of the Q-matrix of a test as a fundamental condition to guarantee a reliable estimate of an examinee's attribute profile has only recently been realized by researchers. In fact, inspection of extant assessments based on the cognitive diagnosis framework often revealed that, in hindsight, the Q-matrices used with these tests were not complete—a prominent example is discussed in detail below. More to the point, what is the value of rules for establishing Q-completeness if the damage has already been done?—that is, the test with an incomplete Q-matrix has been in use for quite a while, and examinees have possibly been misdiagnosed? Said differently, the availability of rules for building a complete Q-matrix at the early stages of test development is perhaps at least as desirable as rules for identifying the completeness of a given Q-matrix. Thus, Part II of this article concerns procedures for constructing Q-matrices that are complete. The famous Fraction-Subtraction test problems by K.K. Tatsuoka (1984) are used throughout for illustration.

3.1 The Fraction-Substraction Problems (K.K. Tatsuoka, 1984)

Tatsuoka's Fraction-Subtraction data have been studied extensively in cognitive diagnosis research (e.g., Chen, de la Torre, and Zhang, 2013; Chiu, 2013; Chiu and Douglas, 2013; Chiu and Köhn, 2015; de la Torre, 2008, 2009, 2011; de la Torre and Douglas, 2004, 2008; DeCarlo, 2011, 2012; Henson et al., 2009; Köhn, Chiu, and Brusco, 2015; Mislevy, 1996; C. Tatsuoka, 2002; K.K. Tatsuoka, 1990, 2009). The data set was collected from 536 middle-school students. (As noted by DeCarlo, 2011, p. 24, a number of articles have reported a sample size of 2,144 observations; this is a mistake that arose when the original data were stacked four times.) Various Q-matrices have been devised—post hoc—for the Fraction-Subtraction data. They differ with regard to the specific subsets of items selected for the analysis and which particular attributes were assumed to be required for a correct item response. As DeCarlo (2011, p. 21) observed, "the fraction subtraction data present a case in point about difficulties with respect to specifying a Q-matrix, in that even for these simple items, it is still debatable (after 20 years) as to the correct specification of the Q-matrix; in fact, virtually every researcher who has considered the fraction subtraction data has suggested different modifications of the Q-matrix." For example, Henson et al. (2009) chose a subset of 12 items, for which they devised a Q-matrix involving only three attributes. De la Torre (de la Torre, 2011; de la Torre and Lee, 2013) also studied a subset consisting of 12 items, but involving four attributes. Another Q-matrix variant for the Fraction-Subtraction data used by de la Torre (2008; see also de la Torre and Douglas, 2008; DeCarlo, 2012; Mislevy, 1996) contained 15 items involving five attributes; this Q matrix is shown in Table 1. Based on this Q-matrix, de la Torre (2008) fitted the Fraction-Subtraction data with the DINA model. The resulting item parameter estimates suggested, according to de la Torre, a good fit to the data. But a quick inspection of this Q-matrix shows that it does not contain all K single-attribute items; hence, it cannot be complete for the DINA model. Specifically, given five attributes, there are $2^5 = 32$ pos-

Table 1. Q-Matrix for the Fraction-Subtraction Test Items (de la Torre, 2008; DeCarlo, 2012)

Number	Item	Attributes				
		α_1	α_2	α_3	α_4	α_5
1	3/4 - 3/8	1	0	0	0	0
2	3 1/2 - 2 3/2	1	1	1	1	0
3	6/7 - 4/7	1	0	0	0	0
4	3 - 2 1/5	1	1	1	1	1
5	3 7/8 - 2	1	0	1	0	0
6	4 4/12 - 2 7/12	1	1	1	1	0
7	4 1/3 - 2 4/3	1	1	1	1	0
8	11/8 - 1/8	1	1	0	0	0
9	3 4/5 - 3 2/5	1	0	1	0	0
10	2 - 1/3	1	0	1	1	1
11	4 5/7 - 1 4/7	1	0	1	0	0
12	7 3/5 - 4/5	1	0	1	1	0
13	4 1/10 - 2 8/10	1	1	1	1	0
14	4 - 1 4/3	1	1	1	1	1
15	4 1/3 - 1 5/3	1	1	1	1	0

Note. α_1 = performing the basic fraction-subtraction operation, α_2 = simplifying/reducing, α_3 = separating a whole number from a fraction, α_4 = borrowing 1 from a whole number, α_5 = converting a whole number to a fraction.

sible proficiency classes in total. Computing the expected item response vectors, $\boldsymbol{S}(\boldsymbol{\alpha})$, for the DINA model, however, confirms that this Q-matrix allows merely to identify nine out of the 32 proficiency classes (see Table 2). Hence, $\mathbf{Q}$ is incomplete. Note that the $S_j(\boldsymbol{\alpha})$ were only computed for the seven items with unique—that is, non-redundant—attribute profiles because the remaining eight items did not provide any additional information. (Previous studies of the Fraction-Subtraction data were usually focussed on item parameter estimation and model fit, but not on the assignment of examinees to proficiency classes, which might explain why this problematic aspect of the fraction-subtraction data has barely been discussed in the literature; see, for example, DeCarlo, 2011.)

In summary, none of the various Q-matrices that have been devised for the Fraction-Subtraction data is complete for the DINA model (and neither for the DINO model) because they all lack inclusion of the K single-attribute items. As an aside, these Q-matrices, however, all conform to Rule 2—they all have full rank K. In addition, checking all pairs of non-nested $\boldsymbol{\alpha}$ and $\boldsymbol{\alpha}^*$ for indeterminate constellations among the coefficients of the associated $\boldsymbol{S}(\boldsymbol{\alpha})$ and $\boldsymbol{S}(\boldsymbol{\alpha}^*)$ establishes completeness of these Q-matrices for all other DCMs. Hence, the Fraction-Subtraction data might be more suitable for a main effects model or a model containing main effects and interaction terms to guarantee the identifiability of all proficiency classes. In fact, Henson et al. (2009) fitted the subset of the Fraction-Subtraction data with the

Table 2. Identifiable Proficiency Classes for the DINA Model Based on the Q-matrix of K. K. Tatsuoka's (1984) Fraction-Subtraction Test Items Shown in Table 1 (de la Torre, 2008; DeCarlo, 2012)

$\boldsymbol{\alpha} = (\alpha_1\, \alpha_2\, \ldots\, \alpha_5)'$	$\mathbf{q}_j$						
	(10000)	(10100)	(10110)	(10111)	(11000)	(11110)	(11111)
	$S_j(\boldsymbol{\alpha})$						
$\boldsymbol{\alpha}_1 = (00000)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_2 = (00001)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_3 = (00010)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_4 = (00011)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_5 = (00100)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_6 = (00101)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_7 = (00110)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_8 = (00111)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_9 = (01000)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{10} = (01001)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{11} = (01010)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{12} = (01011)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{13} = (01100)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{14} = (01101)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{15} = (01110)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{16} = (01111)'$	β_0	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{17} = (10000)'$	$\beta_0 + \beta_1$	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{18} = (10001)'$	$\beta_0 + \beta_1$	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{19} = (10010)'$	$\beta_0 + \beta_1$	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{20} = (10011)'$	$\beta_0 + \beta_1$	β_0	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{21} = (10100)'$	$\beta_0 + \beta_1$	$\beta_0 + \beta_{13}$	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{22} = (10101)'$	$\beta_0 + \beta_1$	$\beta_0 + \beta_{13}$	β_0	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{23} = (10110)'$	$\beta_0 + \beta_1$	$\beta_0 + \beta_{13}$	$\beta_0 + \beta_{134}$	β_0	β_0	β_0	β_0
$\boldsymbol{\alpha}_{24} = (10111)'$	$\beta_0 + \beta_1$	$\beta_0 + \beta_{13}$	$\beta_0 + \beta_{134}$	$\beta_0 + \beta_{1345}$	β_0	β_0	β_0
$\boldsymbol{\alpha}_{25} = (11000)'$	$\beta_0 + \beta_1$	β_0	β_0	β_0	$\beta_0 + \beta_{12}$	β_0	β_0
$\boldsymbol{\alpha}_{26} = (11001)'$	$\beta_0 + \beta_1$	β_0	β_0	β_0	$\beta_0 + \beta_{12}$	β_0	β_0
$\boldsymbol{\alpha}_{27} = (11010)'$	$\beta_0 + \beta_1$	β_0	β_0	β_0	$\beta_0 + \beta_{12}$	β_0	β_0
$\boldsymbol{\alpha}_{28} = (11011)'$	$\beta_0 + \beta_1$	β_0	β_0	β_0	$\beta_0 + \beta_{12}$	β_0	β_0
$\boldsymbol{\alpha}_{29} = (11100)'$	$\beta_0 + \beta_1$	$\beta_0 + \beta_{13}$	β_0	β_0	$\beta_0 + \beta_{12}$	β_0	β_0
$\boldsymbol{\alpha}_{30} = (11101)'$	$\beta_0 + \beta_1$	$\beta_0 + \beta_{13}$	β_0	β_0	$\beta_0 + \beta_{12}$	β_0	β_0
$\boldsymbol{\alpha}_{31} = (11110)'$	$\beta_0 + \beta_1$	$\beta_0 + \beta_{13}$	$\beta_0 + \beta_{134}$	β_0	$\beta_0 + \beta_{12}$	$\beta_0 + \beta_{1234}$	β_0
$\boldsymbol{\alpha}_{32} = (11111)'$	$\beta_0 + \beta_1$	$\beta_0 + \beta_{13}$	$\beta_0 + \beta_{134}$	$\beta_0 + \beta_{1345}$	$\beta_0 + \beta_{12}$	$\beta_0 + \beta_{1234}$	$\beta_0 + \beta_{12345}$

Note. Identifiable proficiency classes—that is, $\boldsymbol{\alpha} \neq \boldsymbol{\alpha}^* \Rightarrow \boldsymbol{S}(\boldsymbol{\alpha}) \neq \boldsymbol{S}(\boldsymbol{\alpha})^*$—are separated by horizontal lines.

LCDM. Recently, de la Torre (2014) suggested that the Reduced Reparameterized Unified Model (Reduced RUM; Hartz, 2002; Hartz and Roussos, 2008) might provide a good fit to the Fraction-Substraction data.

3.2 Building a Complete Q-Matrix

As completeness of a Q-matrix can only be determined in reference to a particular DCM, two basic scenarios must be distinguished. First, if the DINA model or the DINO model is postulated to underlie the test, then the construction of a complete Q-matrix is straightforward: a test developer just needs to include all K different single-attribute items (Rule 1). Second, if DCMs other than the DINA model or the DINO model are assumed to underlie the test, then constructing a complete Q-matrix is more involved; any effort should first aim at compiling a $K \times K$ candidate Q-matrix that is of full rank (as a necessary condition for completeness). Hence, the row vectors (i.e., the item q-vectors) of this candidate Q-matrix must be linearly independent. However, compiling a set of linearly independent q-vectors is not trivial; especially, when K is large. But tools from linear algebra can help; the following stages are recommended:

(1) Put together a $K \times K$ candidate Q-matrix $\mathbf{Q}_c$ consisting of just $J = K$ items. Check whether $\mathbf{Q}_c$ contains identical q-vectors. (Recall that each duplicate q-vector reduces the rank of $\mathbf{Q}_c$ by 1.) Replace the duplicate q-vectors by alternative items until all q-vectors are distinct. (Note that the `R` function `unique()` can be used to check whether all the q-vectors are distinct.)

(2) Check whether every attribute is measured. If there are k attributes that have not yet been considered, then add k items so that each of the attributes is measured by at least one item.

(3) Check the rank of the resulting candidate Q-matrix (Rule 2). (The `R` function `rankMatrix()` in the package `Matrix` can be used to find the rank of a matrix.)

(4) If $\mathbf{Q}_c$ is not of full rank K, then its item attribute profiles are linearly dependent—that is, one or more item attribute profiles can be expressed as a linear combination of the remaining item attribute profiles. Usually, the difficult part is to determine which specific item attribute profiles are linearly dependent.

Transforming a matrix to reduced row echelon form is a commonly used device in linear algebra for detecting patterns of linear dependence among the columns or rows of a matrix. Thus, the reduced row echelon form of a rank-deficient Q-matrix can be used to obtain diagnostic information on how to remedy the linear-dependence issue of the candidate Q-matrix in question.

In general, a matrix is said to be in *row echelon form* (a) if the first nonzero entry in each row is 1; (b) if row r does not consist entirely of zeros, then the number of leading zero entries in row $r+1$ is greater than the number of leading zero entries in row r; (c) if there are rows with all entries equal to zero, then these rows are below those rows having nonzero entries. A matrix is said to be in *reduced row echelon form* (a) if the matrix is in row echelon form; (b) if the first nonzero entry in each row is the only nonzero entry in its column. (For further details, consult Leon, 1998.—Note that the R function `echelon()` in the package `editrules` can be used to find the reduced row echelon form of a matrix.)

Recall that the candidate Q-matrix $\mathbf{Q}_c$ is of $K \times K$; hence, if $\mathbf{Q}_c$ is of full rank, then its reduced row echelon form displays a "staircase" pattern of ones along the main diagonal. If $\mathbf{Q}_c$ is rank deficient, then the staircase pattern is disrupted such that in certain rows the entry of one has been shifted to the right. The columns that have been "shifted over" point at attributes that are confounded—their individual contributions cannot be separated from other attributes due to the particular composition of the item attribute vectors in $\mathbf{Q}_c$. Hence, items must be added to $\mathbf{Q}_c$ that specifically target the confounded attributes. (Because longer tests are more reliable, adding items should typically be preferred over removing/replacing items. How to use the reduced row echelon form for identifying confounded attributes in $\mathbf{Q}_c$ is described in detail in the example below.)

(5) Repeat (2), (3), and (4) until the rank of the candidate Q-matrix is K. Note that adding further items to $\mathbf{Q}_c$—as might be required by the domain of the test in question—is not going to compromise the completeness of the Q-matrix as long as the rank K candidate $\mathbf{Q}_c$ obtained in (4) is retained.

As an illustration of these stages, consider constructing a complete Q-matrix based on a pool of 20 fraction-subtraction items that involve eight attributes. They are presented in Table 3 (these items have been used before by Chen et al., 2013; Chiu, 2013; Chiu and Douglas, 2013; de la Torre and Douglas, 2004; DeCarlo, 2011).Two scenarios are described in detail where the specific properties of the $K \times K$ Q-matrices initially chosen from the total set of items in Table 3 called for different strategies of working through the four building stages.

3.3 Scenario 1: The Initial Candidate Q-matrix Is Not of Full Rank

The initial candidate Q-matrix $\mathbf{Q}_c$ was formed by the first eight items in Table 3:

Number	Item	Attributes							
		α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8
1	5/3 - 3/4	0	0	0	1	0	1	1	0
2	3/4 - 3/8	0	0	0	1	0	0	1	0
3	5/6 - 1/9	0	0	0	1	0	0	1	0
4	3 1/2 - 2 3/2	0	1	1	0	1	0	1	0
5	4 3/5 - 3 4/10	0	1	0	1	0	0	1	1
6	6/7 - 4/7	0	0	0	0	0	0	1	0
7	3 - 4/7	1	1	0	0	0	0	1	0
8	2/3 - 2/3	0	0	0	0	0	0	1	0

Table 3. 20 Fraction-Subtraction Test Items and Their Attribute Profiles (de la Torre and Douglas, 2004; DeCarlo, 2011)

Number	Item	Attributes							
		α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8
1	5/3 - 3/4	0	0	0	1	0	1	1	0
2	3/4 - 3/8	0	0	0	1	0	0	1	0
3	5/6 - 1/9	0	0	0	1	0	0	1	0
4	3 1/2 - 2 3/2	0	1	1	0	1	0	1	0
5	4 3/5 - 3 4/10	0	1	0	1	0	0	1	1
6	6/7 - 4/7	0	0	0	0	0	0	1	0
7	3 - 4/7	1	1	0	0	0	0	1	0
8	2/3 - 2/3	0	0	0	0	0	0	1	0
9	3 7/8 - 2	0	1	0	0	0	0	0	0
10	4 4/12 - 2 7/12	0	1	0	0	1	0	1	1
11	4 1/3 - 2 4/3	0	1	0	0	1	0	1	0
12	1 1/8 - 1/8	0	0	0	0	0	0	1	1
13	3 3/8 - 2 5/6	0	1	0	1	1	0	1	0
14	3 4/5 - 3 2/5	0	1	0	0	0	0	1	0
15	2 - 1/3	1	0	0	0	0	0	1	0
16	4 5/7 - 1 4/7	0	1	0	0	0	0	1	0
17	7 3/5 - 4/5	0	1	0	0	1	0	1	0
18	4 1/10 - 2 8/10	0	1	0	0	1	1	1	0
19	4 - 1 4/3	1	1	1	0	1	0	1	0
20	4 1/3 - 1 5/3	0	1	1	0	1	0	1	0

Note. α_1 = convert a whole number to a fraction, α_2 = separate a whole number from a fraction, α_3 = simplify before subtracting, α_4 = find a common denominator, α_5 = borrow from whole number part, α_6 = column borrow to subtract the second numerator from the first, α_7 = subtract numerators, α_8 = reduce answers to simplest form.

There were only six unique q-vectors, as could be easily verified by the R function `unique()`; Items 2 and 3 were replications. $\mathbf{Q}_c$ was rank deficient: its rank was six. At least two items had to be added to make $\mathbf{Q}_c$ of full rank $K = 8$. The crucial question was which items to pick? Hence, $\mathbf{Q}_c$ was transformed to reduced row echelon form:

Number	Item	Attributes							
		α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8
1	5/3 - 3/4	1	0	0	0	0	0	0	-1
2	3/4 - 3/8	0	1	0	0	0	0	0	1
3	5/6 - 1/9	0	0	1	0	1	0	0	-1
4	3 1/2 - 2 3/2	0	0	0	1	0	0	0	0
5	4 3/5 - 3 4/10	0	0	0	0	0	1	0	0
6	6/7 - 4/7	0	0	0	0	0	0	1	0
7	3 - 4/7	0	0	0	0	0	0	0	0
8	2/3 - 2/3	0	0	0	0	0	0	0	0

Two disruptions of the staircase pattern occured: in row $\mathbf{q}_5$ at column α_5, and in row $\mathbf{q}_7$ at column α_8. The disruptions point at attributes that are confounded: α_5 and α_8. Their individual contributions cannot be separated from other attributes due to the particular composition of the item attribute vectors in $\mathbf{Q}_c$. Hence, items must be added to $\mathbf{Q}_c$ that specifically target the confounded attributes. So, how to "repair" the staircase pattern?—which is equivalent to increasing the rank of $\mathbf{Q}_c$ from six to $K = 8$. A straightforward fix would have been to add those items to $\mathbf{Q}_c$ that would restore entries of ones to the "gaps" at the row/column locations where the disruption of the staircase pattern occurred. Obvious choices, therefore, would have been the two single-attribute items having q-vectors $\mathbf{e}_5 = (00001000)$ and $\mathbf{e}_8 = (00000001)$. Then, $\mathbf{Q}_c$ would have been of full rank, with the reduced row echelon form equal to:

Number	Item	Attributes							
		α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8
1	5/3 - 3/4	1	0	0	0	0	0	0	0
2	3/4 - 3/8	0	1	0	0	0	0	0	0
3	5/6 - 1/9	0	0	1	0	0	0	0	0
4	3 1/2 - 2 3/2	0	0	0	1	0	0	0	0
5	4 3/5 - 3 4/10	0	0	0	0	1	0	0	0
6	6/7 - 4/7	0	0	0	0	0	1	0	0
7	3 - 4/7	0	0	0	0	0	0	1	0
8	2/3 - 2/3	0	0	0	0	0	0	0	1
9		0	0	0	0	0	0	0	0
10		0	0	0	0	0	0	0	0

Unfortunately, the two single-attribute items, $\mathbf{e}_5$ and $\mathbf{e}_8$, were not available in the collection of items in Table 3. So, what to do?

The second row of the reduced row echelon form of $\mathbf{Q}_c$—which is repeated here for convenience—showed that if the "1" in column α_2 could be eliminated, then this row would have the first nonzero entry in column α_8—exactly what was needed:

Number	Item	Attributes							
		α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8
2	3/4 - 3/8	0	1	0	0	0	0	0	1

As it happened, Item 9 in Table 3 is a single-attribute item—it was a perfect match: $\mathbf{q}_9 = (01000000)$. Thus, adding Item 9 to $\mathbf{Q}_c$ ensured that the first

"1" in the seventh row of the reduced row echelon form occurred in column α_8. (Recall that the disruption in row $\mathbf{q}_5$ at column α_5 of the reduced row echelon form has not yet been fixed.)

Next, look at the third row of the reduced row echelon form of $\mathbf{Q}_c$—what about α_5?

Number	Item	Attributes							
		α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8
3	5/6 - 1/9	0	0	1	0	1	0	0	-1

An item measuring α_5 was needed to eliminate the "1" in column α_5 of the third row. Among the remaining 12 items in Table 3 that had not yet been used in $\mathbf{Q}_c$, seven items required α_5. In choosing simply the first of those seven, Item 10, the candidate Q-matrix $\mathbf{Q}_c$ became:

Number	Item	Attributes							
		α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8
1	5/3 - 3/4	0	0	0	1	0	1	1	0
2	3/4 - 3/8	0	0	0	1	0	0	1	0
3	5/6 - 1/9	0	0	0	1	0	0	1	0
4	3 1/2 - 2 3/2	0	1	1	0	1	0	1	0
5	4 3/5 - 3 4/10	0	1	0	1	0	0	1	1
6	6/7 - 4/7	0	0	0	0	0	0	1	0
7	3 - 4/7	1	1	0	0	0	0	1	0
8	2/3 - 2/3	0	0	0	0	0	0	1	0
9	3 7/8 - 2	0	1	0	0	0	0	0	0
10	4 4/12 - 2 7/12	0	1	0	0	1	0	1	1

The reduced row echelon form of this candidate Q-matrix had a perfect staircase pattern of ones along the main diagonal of the $K \times K$ submatrix formed by the first K rows, which confirmed that $\mathbf{Q}_c$ after the addition of Items 9 and 10 was of full rank $K = 8$.

As a final step, for all pairs of non-nested $\boldsymbol{\alpha}$ and $\boldsymbol{\alpha}^*$, the associated expected item response vectors, $\boldsymbol{S}(\boldsymbol{\alpha})$ and $\boldsymbol{S}(\boldsymbol{\alpha}^*)$, had to be checked for indeterminate constellations of the β-coefficients that might have rendered $\mathbf{Q}_c$ incomplete despite having full rank. For $K = 8$ attributes, there are 26,335 pairs of non-nested $\boldsymbol{\alpha}$ and $\boldsymbol{\alpha}^*$ that needed to be inspected. A routine written in `R` called `Q.check` performs a comprehensive completeness check of any candidate Q-matrix, including also the inspection of all pairs of non-nested $\boldsymbol{\alpha}$ and $\boldsymbol{\alpha}^*$. (A beta-version of `Q.check` is available upon request.) `Q.check` confirmed that $\mathbf{Q}_c$ was complete. Additional items could then be added to $\mathbf{Q}_c$ according to a test developer's specific objectives.

3.4 Scenario 2: The Initial Candidate Q-Matrix Is of Full Rank, But Fails the Nestedness Check

In this scenario, the initial candidate Q-matrix was formed by Items 13 to 20 in Table 3:

Number	Item	Attributes							
		α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8
13	$33/8 - 25/6$	0	1	0	1	1	0	1	0
14	$34/5 - 32/5$	0	1	0	0	0	0	1	0
15	$2 - 1/3$	1	0	0	0	0	0	1	0
16	$45/7 - 14/7$	0	1	0	0	0	0	1	0
17	$73/5 - 4/5$	0	1	0	0	1	0	1	0
18	$41/10 - 28/10$	0	1	0	0	1	1	1	0
19	$4 - 14/3$	1	1	1	0	1	0	1	0
20	$41/3 - 15/3$	0	1	1	0	1	0	1	0

The `R` function `unique()` showed that only seven of the eight q-vectors were distinct; $\mathbf{q}_{14}$ and $\mathbf{q}_{16}$ were identical. Also, attribute α_8 was not measured by any item. Hence, Item 12, which required mastery of α_8, was added to $\mathbf{Q}_c$: $\mathbf{q}_{12} = (00000011)$. The candidate Q-matrix containing Items 12 to 20 was now of full rank $K = 8$. Hence, again, as a final step, the expected item response vectors, $\boldsymbol{S}(\boldsymbol{\alpha})$ and $\boldsymbol{S}(\boldsymbol{\alpha}^*)$, had to be checked for indeterminate constellations of the β-coefficients for all associated pairs of non-nested proficiency class attribute vectors. As a technical detail, note that `Q.check` does not perform an exhaustive assessment of all pairs of non-nested attribute profiles. Rather, `Q.check` terminates when the first pair is encountered with indeterminate β-coefficients in the associated $\boldsymbol{S}(\boldsymbol{\alpha})$ and $\boldsymbol{S}(\boldsymbol{\alpha}^*)$. Among the 26,335 pairs of non-nested $\boldsymbol{\alpha}$ and $\boldsymbol{\alpha}^*$, $\boldsymbol{\alpha}_8$ and $\boldsymbol{\alpha}_{43}$ was the first pair that was flagged for indeterminate β-coefficients. Computing $\boldsymbol{S}(\boldsymbol{\alpha})$ and $\boldsymbol{S}(\boldsymbol{\alpha}^*)$ using the GDM confirmed the diagnosis by `Q.check`:

$\boldsymbol{\alpha} = (\alpha_1\ \alpha_2 \ldots \alpha_8)'$	$\mathbf{q}_{12}$	$\mathbf{q}_{13}$
	(00000011)	(01011010)
	$S_{12}(\boldsymbol{\alpha})$	$S_{13}(\boldsymbol{\alpha})$
$\boldsymbol{\alpha}_8 = (00000010)'$	$\beta_{12(0)} + \beta_{12(7)}$	$\beta_{13(0)} + \beta_{13(7)}$
$\boldsymbol{\alpha}_{43} = (11000001)'$	$\beta_{12(0)} + \beta_{12(8)}$	$\beta_{13(0)} + \beta_{13(2)}$
	$\mathbf{q}_{17}$	$\mathbf{q}_{18}$
	(01001010)	(01001110)
	$S_{17}(\boldsymbol{\alpha})$	$S_{18}(\boldsymbol{\alpha})$
$\boldsymbol{\alpha}_8 = (00000010)'$	$\beta_{17(0)} + \beta_{17(7)}$	$\beta_{18(0)} + \beta_{18(7)}$
$\boldsymbol{\alpha}_{43} = (11000001)'$	$\beta_{17(0)} + \beta_{17(2)}$	$\beta_{18(0)} + \beta_{18(2)}$

$\boldsymbol{\alpha} = (\alpha_1\, \alpha_2 \ldots \alpha_8)'$	$\mathbf{q}_{14}$	$\mathbf{q}_{15}$
	(01000010)	(10000010)
	$S_{14}(\boldsymbol{\alpha})$	$S_{15}(\boldsymbol{\alpha})$
$\boldsymbol{\alpha}_8 = (00000010)'$	$\beta_{14(0)} + \beta_{14(7)}$	$\beta_{15(0)} + \beta_{15(7)}$
$\boldsymbol{\alpha}_{43} = (11000001)'$	$\beta_{14(0)} + \beta_{14(2)}$	$\beta_{15(0)} + \beta_{15(1)}$
	$\mathbf{q}_{19}$	$\mathbf{q}_{20}$
	(11101010)	(01101010)
	$S_{19}(\boldsymbol{\alpha})$	$S_{20}(\boldsymbol{\alpha})$
$\boldsymbol{\alpha}_8 = (00000010)'$	$\beta_{19(0)} + \beta_{19(7)}$	$\beta_{20(0)} + \beta_{20(7)}$
$\boldsymbol{\alpha}_{43} = (11000001)'$	$\beta_{19(0)} + \beta_{19(1)} + \beta_{19(2)}$	$\beta_{20(0)} + \beta_{20(2)}$

Unless additional constraints on the β-coefficients would be imposed (e.g., $\beta_{12(7)} \neq \beta_{12(8)}$, or $\beta_{19(1)} + \beta_{19(2)} \neq \beta_{19(7)}$) completeness of $\mathbf{Q}_c$ would not be guaranteed—in other words, the candidate Q-matrix failed the "nestedness" check.

Thus, more items needed to be added to $\mathbf{Q}_c$ to eliminate the indeterminacy among the β-coefficients of $\boldsymbol{S}(\boldsymbol{\alpha}_8)$ and $\boldsymbol{S}(\boldsymbol{\alpha}_{43})$. The most efficient strategy—although not the only one—was to add an item that had a q-vector equal either to $\boldsymbol{\alpha}_8$ or $\boldsymbol{\alpha}_{43}$—provided such items were available. In fact, among the items not yet selected for $\mathbf{Q}_c$, Item 8 had an attribute profile that was identical to that of proficiency class $\boldsymbol{\alpha}_8$: $\mathbf{q}_8 = (00000010)$. Computing the expected response profile S_8 for $\boldsymbol{\alpha}_8$ and $\boldsymbol{\alpha}_{43}$ confirmed that now the two proficiency classes were well distinguishable:

$\boldsymbol{\alpha} = (\alpha_1\, \alpha_2 \ldots \alpha_8)'$	$\mathbf{q}_8$
	(00000010)
	$S_8(\boldsymbol{\alpha})$
$\boldsymbol{\alpha}_8 = (00000010)'$	$\beta_{8(0)} + \beta_{8(7)}$
$\boldsymbol{\alpha}_{43} = (11000001)'$	$\beta_{8(0)}$

After Item 8 had been added to $\mathbf{Q}_c$, the candidate Q-matrix was run again through `Q.check`; now, it was complete:

Number	Item	Attributes							
		α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8
8	$2/3 - 2/3$	0	0	0	0	0	0	1	0
12	$11/8 - 1/8$	0	0	0	0	0	0	1	1
13	$33/8 - 25/6$	0	1	0	1	1	0	1	0
14	$34/5 - 32/5$	0	1	0	0	0	0	1	0
15	$2 - 1/3$	1	0	0	0	0	0	1	0
16	$45/7 - 14/7$	0	1	0	0	0	0	1	0
17	$73/5 - 4/5$	0	1	0	0	1	0	1	0
18	$41/10 - 28/10$	0	1	0	0	1	1	1	0
19	$4 - 14/3$	1	1	1	0	1	0	1	0
20	$41/3 - 15/3$	0	1	1	0	1	0	1	0

But if Item 8 had not been among the candidate items? An alternative solution would have been to construct an item with an attribute profile that was nested within the attribute profile of $\boldsymbol{\alpha}_{43}$—say, $\mathbf{q}^* = (11000000)$. Then, as the expected response profiles showed, the two proficiency classes would have been well distinguishable:

$\boldsymbol{\alpha} = (\alpha_1\ \alpha_2\ \ldots\ \alpha_8)'$	$\mathbf{q}^*$
	(11000000)
	$S^*(\boldsymbol{\alpha})$
$\boldsymbol{\alpha}_8 = (00000010)'$	β_0^*
$\boldsymbol{\alpha}_{43} = (11000001)'$	$\beta_0^* + \beta_1^*$

Recall that, as long as there is a single S_j such that $S_j(\boldsymbol{\alpha}) < S_j(\boldsymbol{\alpha}^*)$ or $S_j(\boldsymbol{\alpha}) > S_j(\boldsymbol{\alpha}^*)$, $\boldsymbol{S}(\boldsymbol{\alpha}) \neq \boldsymbol{S}(\boldsymbol{\alpha}^*)$.

4. Discussion

A test with an incomplete Q-matrix does not allow for the identification of all proficiency classes, which, in turn, causes examinees to be misclassified. In practice, such false ability assessment can have grave consequences for the test taker as well as the test administrator. Related work by Maddison and Bradshaw (2015) and by Xu and Zhang (2016) showed that completeness of the Q-matrix is also an indispensable requirement for the identifiability (and estimability) of the item parameters. (Madison, Bradshaw, and Hollingsworth [2014] developed a web-based device—located at http://www.lainebradshaw.com/qpower/ —that allows researchers to assess beforehand to what extent their candidate Q-matrix might be at risk of failing to ensure estimable item parameters.) In this article, a four-stage sequential procedure was presented for constructing Q-matrices that are guaranteed to be complete. First, check wether the cadidate Q-matrix $\mathbf{Q}_c$ contains all single-attribute q-vectors; if so, then $\mathbf{Q}_c$ is complete. Second, if any of the single-attribute item profiles is missing from $\mathbf{Q}_c$, and the data are fitted by the DINA model or the DINO model, then $\mathbf{Q}_c$ is not complete. Third, if the fitted model is neither the DINA nor the DINO, then check the rank of $\mathbf{Q}_c$; if it is less than K, then $\mathbf{Q}_c$ is incomplete. Fourth, if $\mathbf{Q}_c$ has full rank K, then check the expected item response profiles of all non-nested pairs $\boldsymbol{\alpha}$ and $\boldsymbol{\alpha}^*$ for indeterminate β-coefficients. Note that the steps of the four-stage procedure provide a "built-in" safe-guard against unintentionally constructing an incomplete Q-matrix even if the true DCM underlying a test is not known with certainty—which, in practice, is likely the rule than the exception. In conclusion, two closely related questions should be briefly addressed.

The first concerns the effect on Q-completeness of sequential model comparison strategies, as they might be used for identifying the best fitting model among candidate DCMs when information on the true DCM underlying the data is not available (e.g., Chen et al., 2013; de la Torre and Lee, 2013; Rupp et al., 2010; Templin and Bradshaw, 2014). Recall that completeness of the Q-matrix depends on the chosen model. Model comparison strategies, however, focus on identifying the best fitting model and usually ignore whether completeness of the Q-matrix is secured for the various candidate models. But completeness of the Q-matrix may not necessarily be preserved when the model supposedly underlying the data is modified. Thus, the recommendation is to monitor Q-completeness during the model-finding process.

The second question refers to situations when a test contains items that do not just conform to a single DCM, but presumably to a mix of multiple DCMs. How can the four-stage procedure be used to secure that the Q-matrix is still complete, given such uncertainties?

As a simple example, assume that the items of a test conform to two DCMs, the DINA model and the GDM, involving K attributes. If the Q-matrix contains all K single-attribute items, then its completeness is guaranteed regardless of the specific mix of DCMs in the test. In all other instances, completeness of the Q-matrix is no longer given for the DINA model. However, if $\mathbf{Q}_c$ contains a $K \times K$ submatrix that has full rank, then the Q-matrix can still be complete for the GDM provided indeterminacies of the β-coefficients in the expected item responses can be ruled out. To illustrate the crucial role of the particular choice of q-vectors in $\mathbf{Q}_c$, consider a test with only three items, $\mathbf{q}_1 = (100)$, $\mathbf{q}_2 = (011)$, and $\mathbf{q}_3 = (001)$, Item 1 conforms to the DINA model, whereas the other two items conform to the GDM. The Q-matrix is of full rank and complete, as the inspection of the expected item responses confirms:

$\boldsymbol{\alpha} = (\alpha_1\ \alpha_2\ \alpha_3)'$	$\mathbf{Q}_c$		
	DINA	GDM	
	$\mathbf{q}_1 = (100)$	$\mathbf{q}_2 = (011)$	$\mathbf{q}_3 = (001)$
	$S_1(\boldsymbol{\alpha})$	$S_2(\boldsymbol{\alpha})$	$S_3(\boldsymbol{\alpha})$
$\boldsymbol{\alpha}_1 = (000)'$	β_{10}	β_{20}	β_{30}
$\boldsymbol{\alpha}_2 = (001)'$	β_{10}	$\beta_{20} + \beta_{23}$	$\beta_{30} + \beta_{33}$
$\boldsymbol{\alpha}_3 = (010)'$	β_{10}	$\beta_{20} + \beta_{22}$	β_{30}
$\boldsymbol{\alpha}_4 = (011)'$	β_{10}	$\beta_{20} + \beta_{22} + \beta_{23}$	$\beta_{30} + \beta_{33}$
$\boldsymbol{\alpha}_5 = (100)'$	$\beta_{10} + \beta_{11}$	β_{20}	β_{30}
$\boldsymbol{\alpha}_6 = (101)'$	$\beta_{10} + \beta_{11}$	$\beta_{20} + \beta_{23}$	$\beta_{30} + \beta_{33}$
$\boldsymbol{\alpha}_7 = (110)'$	$\beta_{10} + \beta_{11}$	$\beta_{20} + \beta_{22}$	β_{30}
$\boldsymbol{\alpha}_8 = (111)'$	$\beta_{10} + \beta_{11}$	$\beta_{20} + \beta_{22} + \beta_{23}$	$\beta_{30} + \beta_{33}$

Now, assume that Items 1 and 2 conform to the DINA model, but Item 3 to the GDM. The Q-matrix is still of full rank, but no longer complete, as can be demonstrated by calculating the expected item responses (note that the proficiency classes $\boldsymbol{\alpha}_1 = (000)'$ and $\boldsymbol{\alpha}_3 = (010)'$ as well as $\boldsymbol{\alpha}_5 = (100)'$ and $\boldsymbol{\alpha}_7 = (110)'$ cannot be distinguished):

$\boldsymbol{\alpha} = (\alpha_1\ \alpha_2\ \alpha_3)'$	$\mathbf{Q}_c$		
	DINA		GDM
	$\mathbf{q}_1 = (100)$	$\mathbf{q}_2 = (011)$	$\mathbf{q}_3 = (001)$
	$S_1(\boldsymbol{\alpha})$	$S_2(\boldsymbol{\alpha})$	$S_3(\boldsymbol{\alpha})$
$\boldsymbol{\alpha}_1 = (000)'$	β_{10}	β_{20}	β_{30}
$\boldsymbol{\alpha}_2 = (001)'$	β_{10}	β_{20}	$\beta_{30} + \beta_{33}$
$\boldsymbol{\alpha}_3 = (010)'$	β_{10}	β_{20}	β_{30}
$\boldsymbol{\alpha}_4 = (011)'$	β_{10}	$\beta_{20} + \beta_{2(23)}$	$\beta_{30} + \beta_{33}$
$\boldsymbol{\alpha}_5 = (100)'$	$\beta_{10} + \beta_{11}$	β_{20}	β_{30}
$\boldsymbol{\alpha}_6 = (101)'$	$\beta_{10} + \beta_{11}$	β_{20}	$\beta_{30} + \beta_{33}$
$\boldsymbol{\alpha}_7 = (110)'$	$\beta_{10} + \beta_{11}$	β_{20}	β_{30}
$\boldsymbol{\alpha}_8 = (111)'$	$\beta_{10} + \beta_{11}$	$\beta_{20} + \beta_{2(23)}$	$\beta_{30} + \beta_{33}$

In conclusion, as this small-scale example suggests, matters become quickly pretty complicated even if just two different DCMs are underlying a test. The general recommendation for tests involving multiple DCMs is to play it safe and include all K single-attribute items in the Q-matrix, which guarantees completeness for any mix of DCMs. If this is not possible—in certain knowledge domains, it is often difficult to devise single-attribute items, not to mention all K of them—then the only option is to monitor the Q-matrix construction closely in applying the four-stage procedure developed here.

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